

Source Data Used in this App

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Below are brief descriptions of the source datasets used in this app. For more details, please refer to the linked original paper.

bulk RNAseq Dataset

General Information

Tabula Muris Senis (bulk Dataset)

<https://tabula-muris-senis.sf.czbiohub.org>

Description

Tabula Muris Senis is a comprehensive transcriptomic cell atlas of mice covering a wide range of ages. The atlas includes both single-cell RNA-seq and bulk RNA-seq data. In this app, only bulk RNA-seq data are used.

Link to the Original Paper

Schaum N, Lehallier B, Hahn O, et al. Ageing hallmarks exhibit organ-specific temporal signatures. Nature. 2020;583(7817):596-602.

[doi:10.1038/s41586-020-2499-y](https://doi.org/10.1038/s41586-020-2499-y)

Sample Information

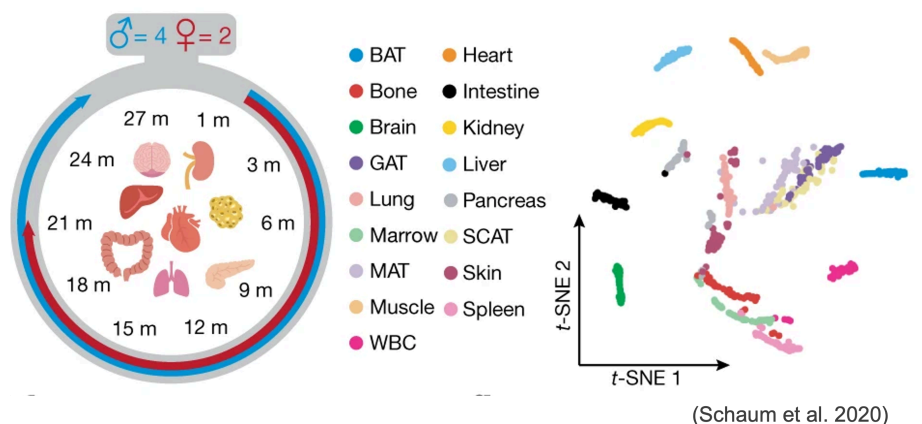
Mouse Strain

C57BL/6JN

Age/Sex Group Composition

- male mice (n = 4; aged 1, 3, 6, 9, 12, 15, 18, 21, 24 and 27 months)
- female mice (n = 2; aged 1, 3, 6, 9, 12, 15, 18 and 21 months)

Note: The numbers above indicate the number of mice; actual sample counts per tissue may be lower due to quality filtering.



Tissue

Samples from 17 tissues are available:

- **BAT**: brown adipose tissue (interscapular depot)
- **Bone** (femurs and tibiae)
- **Brain** (hemibrain)
- **GAT**: gonadal adipose tissue (inguinal depot)
- **Heart**
- **Kidney**
- **Limb Muscle** (tibialis anterior)
- **Liver**
- **Lung**
- **Marrow**: bone marrow
- **MAT**: mesenteric adipose tissue
- **Pancreas**
- **SCAT**: subcutaneous adipose tissue (posterior depot)
- **Skin** (dorsal)
- **Small Intestine** (duodenum)
- **Spleen**
- **WBC**: white blood cells (buffy coat)

Measurement Platform

Illumina NovaSeq 6000 (Mus musculus)

Original Data Repository

Gene Expression Omnibus ([GSE132040](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE132040))

Proteomic Dataset

General Information

Description

Proteomic data were retrieved from the paper by Keele et al. (2023). In this study, the authors performed discovery-based proteomics using tandem mass tags (TMTs) in 10 tissues from 20 C57BL/6J mice, including both sexes and two age groups (8 and 18 months).

Link to the Original Paper

Keele GR, Zhang JG, Szpyt J, et al. Global and tissue-specific aging effects on murine proteomes. Cell Rep. 2023;42(7):112715.

[doi:10.1016/j.celrep.2023.112715](https://doi.org/10.1016/j.celrep.2023.112715)

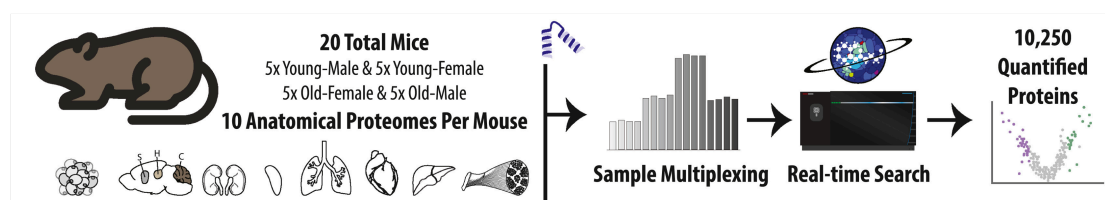
Sample Information

Mouse Strain

C57BL/6J

Age/Sex Group Composition

- 8-month male (N=5)
- 8-month female (N=5)
- 18-month male (N=5)
- 18-month female (N=5)



Tissue

10 tissues were analyzed:

- **Kidney**
- **Lung**
- **Liver**
- **Heart**
- **Skeletal Muscle** (quadriceps)
- **Fat** (inguinal adipose)
- **Hippocampus**
- **Cerebellum**
- **Striatum**

Measurement Platform

Tandem mass tags (TMTs)

Original Data Repository

Fig Share ([DOI link](#))

Metabolomic Dataset

General Information

Description

Metabolomic data were retrieved from the paper by Jankowski et al. (2025). In this study, the authors performed both metabolomics and stable isotope tracing on young (3–6 months) and aged (+21 months) C57BL/6J mice. In this app, only the metabolomic data are used.

Link to the Original Paper

Jankowski CSR, Samarah LZ, MacArthur MR, et al. Aged mice exhibit widespread metabolic changes but preserved major fluxes. Cell Metab. 2025;37(11):2280-2294.e4.

[doi:10.1016/j.cmet.2025.09.009](https://doi.org/10.1016/j.cmet.2025.09.009)

Sample Information

Mouse Strain

C57BL/6J

Age/Sex Group Composition

Each group (young male, young female, aged male, aged female) consisted of 8–10 samples per tissue (Approximately 70% of tissue–sex–age groups had N = 10).

Tissue

15 tissues (including serum) were analyzed:

- Serum
- Brain
- Eye
- Diaphragm
- Heart
- Jejunum
- Kidney
- Liver
- Lung
- Pancreas
- Spleen
- Colon
- Quadriceps
- Soleus
- Skin (Ear)

Measurement Platform

Hydrophilic interaction chromatography (HILIC) coupled with a quadrupole-orbitrap mass spectrometer (QExactive Plus, Exploris 240, or Exploris 480)

Original Data Source

Raw data were obtained from the [Supplementary Table](#) (Table S6) of the original publication. Individual metabolite data are also available at the R/Shiny app developed by the authors: <https://agingmet.princeton.edu/> (last accessed 2025-12-08)